The Latent Geometry of "Meh"

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Abstract

Where does a one-word social response sit inside a language model? We fit twelve signed residual-stream read directions in two Qwen 3.6 checkpoints, using gradients of positive-minus-negative pole-token logits over eight neutral prompts. The 96 held-out landmark utterances and 126-item social atlas contain none of the 144 pole words. On the confirmatory Qwen3.6-35B-A3B run, all twelve landmark means have the intended orientation and ten have positive 95% lower bounds. The centroid of six variants of *meh* has engagement -4.94 calibrated units, with a 95% interval of $[-5.77, -3.73]$, and lies 5.98 units from neutral. Its nearest held-out landmark is discouragement, which fails a frozen boredom-adjacency rule. A descriptive Qwen3.6-27B extension reproduces the twelve-coordinate profile: the two *meh* centroids have cosine 0.951 and agree in sign on every coordinate. Boredom is nearest in the dense model. The nearest label also changes through residual depth in both models. The targeted J-space slice gives *meh* a stable low-engagement direction, while its categorical address depends on model and layer.

1 Introduction

Meh carries more social information than its length suggests. It can signal boredom, discouragement, dismissal, low effort, or amused detachment. Those readings overlap, and a single conversational setting can shift their balance. This makes the word a compact test for whether a residual-space map captures a directional profile or assigns a fixed category.

The Jacobian lens maps intermediate activations into token-indexed directions defined by their average linearized effect on later output logits [2]. Its full J-space is a sparse, nonnegative frame built from those directions. The method can expose concepts that do not appear in the model's output, and its readouts change across layers.

We study twelve output-defined directions chosen to cover a social and affective spectrum. The axes run from warmth to hostility, trust to wariness, affiliation to distance, playfulness to formality, ease to tension, care to indifference, engagement to boredom, patience to annoyance, efficacy to frustration, social safety to hurt or defensiveness, goodwill to resentment, and hope to discouragement. Each direction comes from pole-token gradients. Neutral messages set its location and scale.

The evaluation separates the words used to define the directions from the utterances used to test them. It also records a complete token-by-layer replay, runs three system prompts, and freezes four decisions before the 35B confirmatory run. The 27B dense checkpoint supplies a model-architecture comparison with the same data and analysis.

2 Relation to prior methods

Linear probes can recover interpretable structure from language-model activations, but probe performance alone does not show that the model uses the measured feature [3]. Held-out transfer and intervention tests have provided stronger evidence for linear directions in settings such as factual truth [4]. Tuned lenses learn a separate map from each layer to the output distribution [1]. The Jacobian lens instead uses the downstream derivative, giving the direction a local causal interpretation with respect to output logits [2].

Activation additions use related residual directions to alter generated text [8, 9]. Emotion representations have also been causally linked to output and alignment behavior in Claude Sonnet 4.5 [7]. Our runner only reads activations. It does not intervene or decode responses, so its results establish geometry rather than a causal effect on behavior.

3 Method

3.1 A targeted Jacobian readout

For axis a, let P_a and N_a be the resolved single-token IDs for six positive and six negative pole words. Capitalized and space-prefixed tokenizer forms are included when they remain single tokens. For source residual activation $h_{\ell,t}$ and target-layer logits z, the read direction is

$$v_a = \mathbb{E}_{x,t,t' \geq t} \nabla_{h_{\ell,t}} \left[\frac{1}{|P_a|} \sum_{p \in P_a} z_{p,t'} - \frac{1}{|N_a|} \sum_{n \in N_a} z_{n,t'} \right]. \tag{1}$$

The expectation is estimated over valid positions in eight neutral prompts. The first eight positions are excluded and sequences are capped at 96 tokens. All model weights remain frozen.

For system prompt s and traced layer j, 24 neutral calibration messages give a coordinate-wise center $\mu_{a,s,j}$ and population standard deviation $\sigma_{a,s,j}$. The calibrated coordinate of user token t is

$$c_{a,s,j,t} = \frac{\langle h_{j,t}, v_a \rangle - \mu_{a,s,j}}{\sigma_{a,s,j}}. \tag{2}$$

An utterance coordinate is the mean over its user-content tokens. The token span is recovered by comparing chat templates with an empty and populated user turn, which removes role markers, assistant headers, and template newlines.

This construction is a targeted J-space readout. It does not estimate the full $d \times d$ averaged Jacobian, rank the full vocabulary, or solve the sparse nonnegative decomposition that defines the full J-space [2]. For depth traces, the source-layer matrix $[v_1, \ldots, v_{12}]$ is reused at every sampled layer with a fresh neutral calibration for that layer. Those traces are fixed-direction residual-depth measurements.

3.2 Models and frozen layers

The confirmatory checkpoint is Qwen3.6-35B-A3B revision `995ad96eacd98c81ed38be0c5b274b04031597b0`. It has 35B total parameters, 3B active parameters, a 2,048-dimensional language-model state, and 40 layers [6]. We fit from residual layer 35 to layer 39 and trace layers 3, 7, …, 39.

The descriptive extension uses Qwen3.6-27B revision `6a9e13bd6fc8f0983b9b99948120bc37f49c13e9`. It has 27B parameters, a 5,120-dimensional state, and 64 layers [5]. We fit from layer 55 to 63 and trace layers 3, 7, …, 63. Both checkpoints run in bfloat16 with PyTorch SDPA and the model cache disabled.

3.3 Lexically disjoint evaluation

Each signed axis has four positive and four negative landmark paraphrases, for 96 cases. The social atlas has 21 families with six utterances each. Its *meh* family contains three bare forms ("meh", "meh.", and "Meh.") and three contextual forms ("meh, whatever", "meh, I could take it or leave it", and "meh, next"). The other families cover greetings, gratitude, apology, praise, insult, dismissal, complaint, repeated demands, encouragement, agreement, disagreement, uncertainty, surprise, amusement, farewell, requests, refusal, relief, confusion, and neutral statements.

A case-insensitive whole-word check rejects evaluation text containing any pole word. The frozen inventory has zero collisions. Every case runs under helpful/direct, concise/technical, and conversational/natural system prompts: $222 \times 3 = 666$ passes per model.

3.4 Metrics and decisions

Intervals use 2,000 bootstrap resamples with seed 20260726. Resampling is clustered by utterance variant, with one system prompt drawn inside each selected variant. These intervals measure sensitivity to the fixed prompt inventory.

Four rules were written before the confirmatory run:

1. all twelve target landmark means must be positive, with at least ten positive 95% lower bounds;

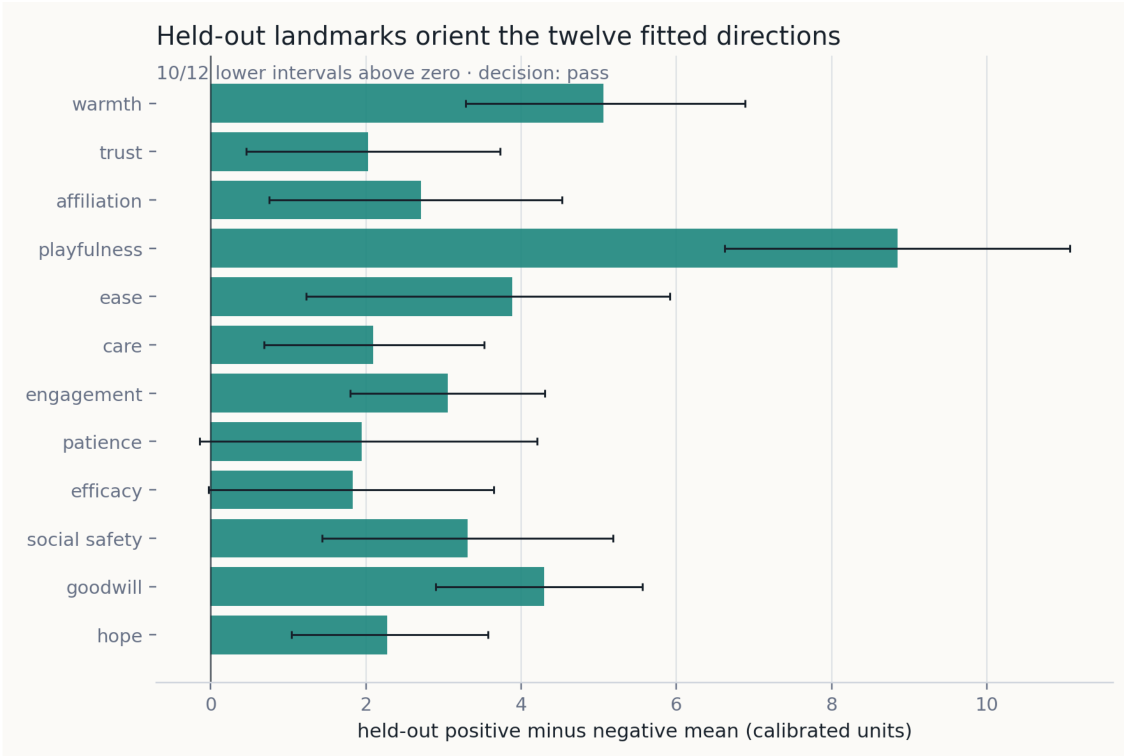

Figure 1: Positive-minus-negative landmark separation on the intended coordinate for Qwen3.6-35B-A3B. Intervals are clustered over the four utterance variants and three system prompts. None of the landmark text contains a pole word.

2. the upper engagement interval for *meh* must be below zero and its distance from neutral at least three calibrated units;
3. boredom must be the nearest landmark overall and under at least two system prompts;
4. the median same-family cosine across system prompts must be at least 0.80.

The 27B comparison and all depth results are descriptive.

4 Results

4.1 Held-out utterances orient the directions

Figure 1 shows the confirmatory landmark separations. All twelve target means are positive; ten lower intervals stay above zero. Patience and efficacy have positive means with lower bounds of -0.139 and -0.029. The 27B extension also has twelve positive means and eleven positive lower bounds; efficacy is the exception at -0.235.

The intended coordinate accounts for a median 51.9% of the full twelve-axis landmark separation vector on 35B and 57.3% on 27B. Off-axis movement is substantial. Mean absolute off-diagonal axis correlation is 0.343 and 0.360, and the first principal component explains 48.8% and 54.1% of atlas-family centroid variance. The radar axes should therefore be read as correlated contrasts rather than orthogonal bins.

4.2 A stable low-engagement profile

Table 1 gives the main measurements. The 35B *meh* centroid has engagement -4.941 with an upper interval of -3.727. Its Euclidean distance from the neutral centroid is 5.976. Both parts of the non-null rule pass. The dense model yields engagement -3.457, upper interval -2.043, and distance 6.194.

Figure 2 shows the full confirmatory coordinate and its contrast with neutral utterances. Relative to neutral, the largest negative shifts are engagement (-3.98) and patience (-2.94). Playfulness and ease move positively.

Table 1: Source-layer results. The 35B column is confirmatory; the 27B column is a descriptive extension.

Measurement	Qwen3.6-35B-A3B	Qwen3.6-27B
Target means above zero	12/12	12/12
Target lower bounds above zero	10/12	11/12
Meh engagement	-4.941	-3.457
Engagement 95% interval	$[-5.770, -3.727]$	$[-4.341, -2.043]$
Meh-neutral distance	5.976	6.194
Nearest landmark	discouragement	boredom
Boredom nearest by system	0/3	3/3
Template cosine median	0.9958	0.9957

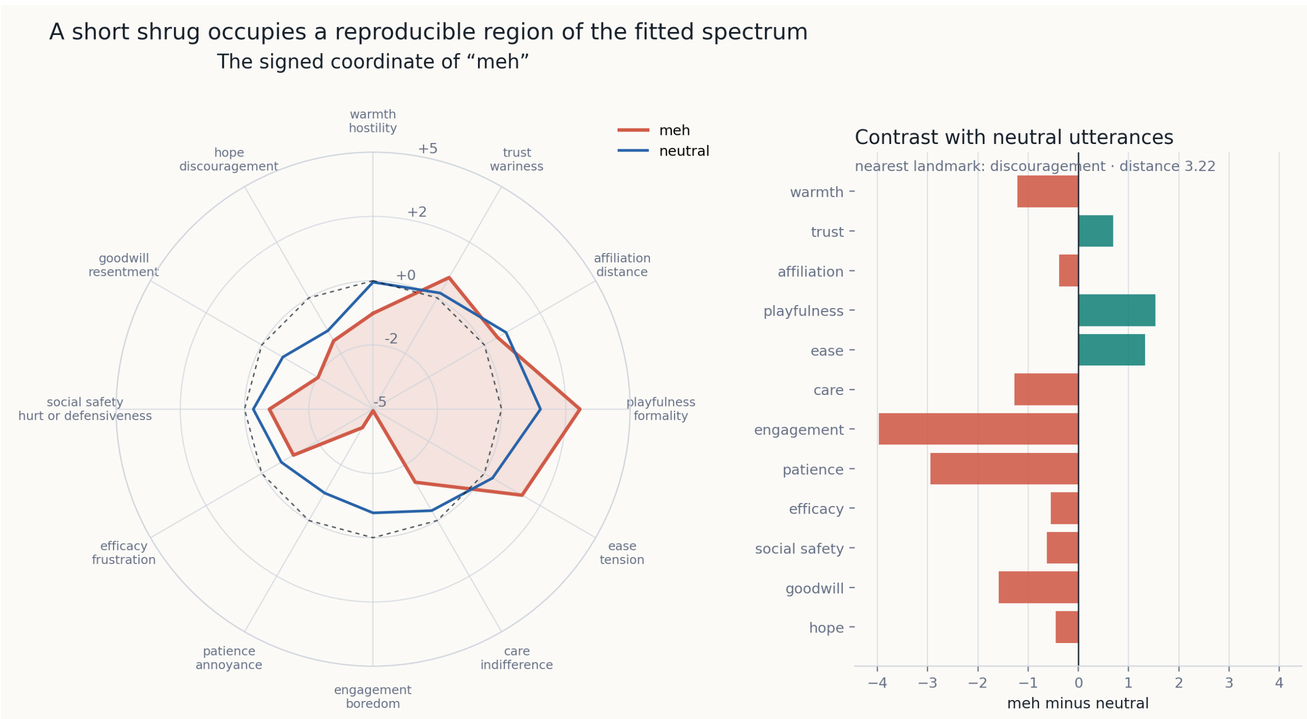

Figure 2: The twelve-coordinate *meh* centroid on Qwen3.6-35B-A3B. The radar compares raw calibrated family centroids; the bars subtract the neutral-family centroid.

Bare and contextual forms point in similar directions, while the contextual forms move farther on engagement and patience.

The cross-model cosine between *meh* centroids is 0.951, with identical signs on all twelve coordinates. Across all 21 matching atlas families, median cosine is 0.955 and mean coordinate-sign agreement is 94.0%. The three system templates also leave family geometry nearly unchanged: their median same-family cosine is 0.9958 on 35B and 0.9957 on 27B.

4.3 The categorical address moves

The frozen boredom-adjacency rule fails on the confirmatory model. Discouragement is nearest at layer 35 with Euclidean distance 3.216, followed by frustration at 3.297 and boredom at 3.720. Discouragement remains first under every system prompt. On the dense model, boredom is nearest overall at 2.654 and under all three prompts.

The fixed-direction depth trace changes the answer again. On 35B, the nearest landmark is playfulness at layers 3 and 7; frustration at 11 and 15; boredom from 19 through 27; frustration at 31; discouragement at the frozen source layer 35; and boredom at the final traced layer 39. On 27B, late layers alternate mainly between boredom and frustration, with boredom nearest at layers 27, 35, 39, 43, 55, 59, and 63.

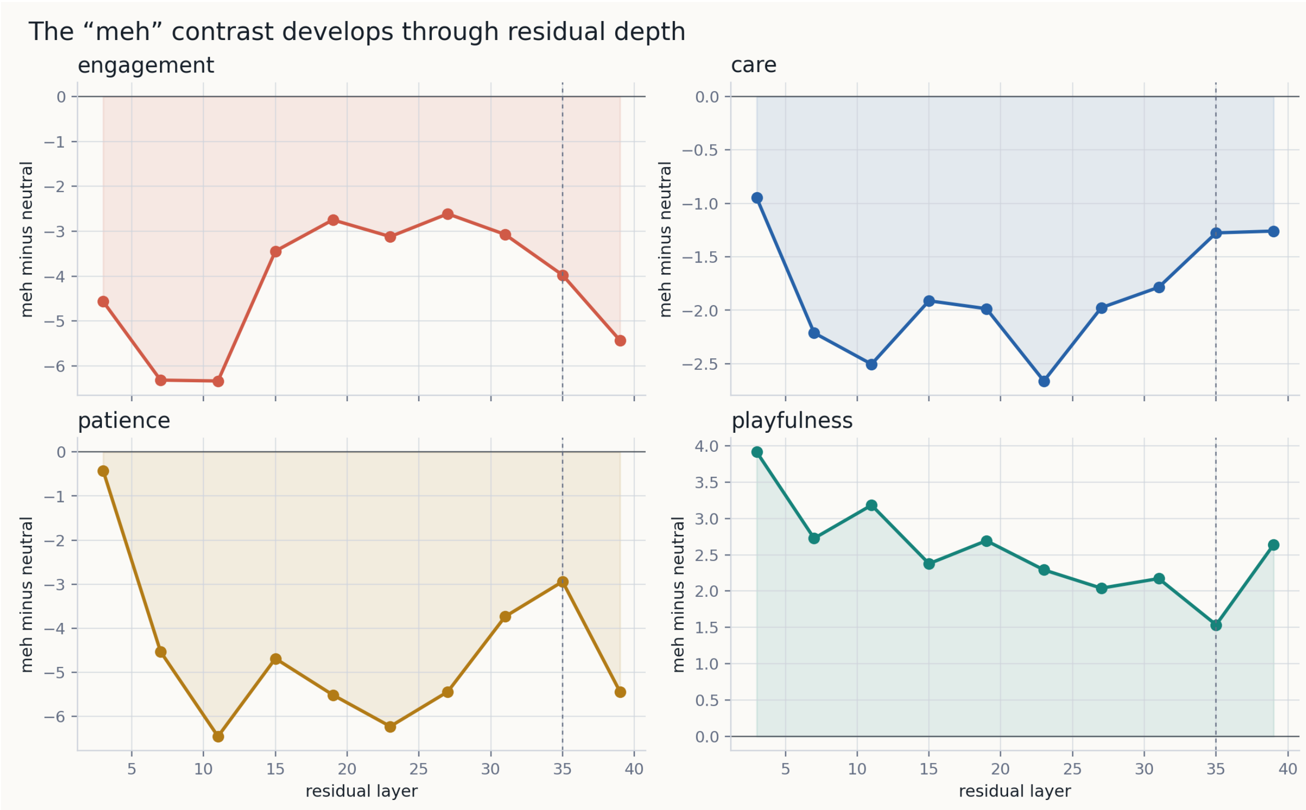

Figure 3: Four coordinates of the *meh*-minus-neutral contrast through Qwen3.6-35B-A3B residual depth. The dashed line marks the fitted source layer. Every layer uses the layer-35 read directions and its own neutral calibration.

The directional profile survives the model comparison better than the nearest category. Low engagement and low patience recur in both checkpoints. A Euclidean winner in twelve correlated coordinates compresses that profile into one atlas label, and small changes in the remaining coordinates can change the winner.

5 Limits and next tests

The corpus is synthetic, English-only, and small. It covers six variants per atlas family, four variants per landmark pole, and three system prompts. The bootstrap intervals describe this inventory rather than a population of users, decoding samples, or checkpoints. Both models come from the same Qwen release, so the 27B result tests architecture and scale inside one model family.

The pole words determine the read directions. Lexical separation prevents a direct word-matching shortcut in evaluation, but another reasonable pole inventory could rotate the slice. Neutral calibration also makes coordinates comparable within this experiment, not universal units. Correlated axes make Euclidean neighbors especially sensitive to the chosen inventory and scale.

The depth trace applies one fitted direction matrix outside its source layer. Independent J-lens fits at every layer would separate genuine representational rotation from mismatch between a layer and the reused read matrix. A broader replication should add independently fitted layers, unrelated model families, more system prompts, and human similarity judgments.

The measurements say nothing about persistent state or model experience. They also do not show whether moving an activation along these directions changes a response. A direct follow-up can intervene along the measured axes and score complete decoded responses, while keeping the lexically disjoint atlas fixed.

6 Reproducibility

The runner is self-contained. It pins both model revisions, the Jacobian-lens implementation, Python packages, corpus hash `a9b52273f23fcff7845e1d3e49bdfc7f34d8262ed1814be2800650348329deb4`, calibration, bootstrap seed, metrics, and decisions. Both paper runs used Python 3.14.3, PyTorch 2.12.0, Transformers 5.12.1, CUDA 13.0, and one NVIDIA RTX PRO 6000. The 35B measurement run took 114.2 seconds with its checkpoint already cached. The 27B run took 670.6 seconds including the initial checkpoint download and reserved at most 50.5 GiB of CUDA memory.

The repository includes the frozen method, source, tests, summaries, raw token measurements, PNG figures, cross-model comparison, and a self-contained HTML replay for each model. Resumable fit states and duplicate event streams are excluded from the publication bundle. The commands are:

```
uv sync --frozen
uv run pytest
uv run jspace-spectrum-paper reproduce --suite full \
  --model qwen36-35b --device cuda \
  --output-dir results/qwen36-35b-confirmatory
uv run jspace-spectrum-paper reproduce --suite full \
  --model qwen36-27b --device cuda \
  --output-dir results/qwen36-27b-extension
```

References

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